



Investigation report

Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads -
ATSB-style final report

Drafted 04/10/2026

INVESTIGATION WORKFLOW SUITE

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Investigation summary

What happened

On 07/07/2026, MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry, was inbound through the Weerona Channel at Kanooka Heads on TF18, the last service of the day. It carried 38 passengers, a master, a deckhand and a kiosk attendant. Under the winter timetable the transit was made in darkness. At 18:07:06 the chart plotter display froze and did not recover. The master continued at about 8 kt while searching for the No. 4 beacon. The heading increased from 195 to 200 degrees and was held, and the vessel left the dredged channel at about 18:09. At 18:12:04, 16 seconds after the echo sounder shallow alarm sounded, it made contact with Boolarong Bank at 6.8 kt. It came to rest 74 m west of the charted centreline and 54 m outside the channel.

There were no serious injuries. One passenger reported a grazed shin and two were treated at the wharf for distress. The starboard hull bottom plating was set up over 4.1 m with a 180 mm split, the No. 2 void flooded to approximately 260 mm, and the starboard propeller, shaft, rudder stock and skeg were damaged. The master stopped both engines and did not attempt to refloat on the falling tide, which very likely reduced the risk to those on board. All passengers were ashore by 19:26. The vessel refloated at 00:20 on 08/07/2026 and was withdrawn from service. The channel was closed to commercial traffic for 19 hours.

What the investigation found

The plotter freeze removed the primary means of position fixing. Continuing at eight knots very likely contributed, because stopping at any point in the 114 seconds before the vessel crossed the channel edge would have kept it in the channel. The operator had no night passage plan for the channel and no hold point for loss of the plotter (Safety Issue). Where the master searched, the No. 4 beacon was very likely not visible, screened by mangroves on Weerona Islet whose clearing had been deferred as unfunded at three works rounds (Safety Issue). The 0.9 m shallow alarm setting gave 58 m of warning against a 92 m crash stop distance.

The winter timetable change moved TF18 into darkness without reopening the route risk assessment, which restricted the channel to daylight (Safety Issue). The plotter fault was passed on verbally at the change of master, as operator procedures permitted (Safety Issue), so firmware addressing it was never applied. Each likely contributed, although a defect entry on the day would not alone have stopped TF18 sailing.

The beacon's solar conversion halved its nominal range without operators being told, and the waterway's aids carried no availability or notification duties (Safety Issues). Both likely contributed, on weaker evidence: the light was screened where the master searched regardless of its strength, and nothing in the code prevented the Authority from notifying operators.

Whether floodlighting behind the beacon affected detection could not be determined. The forward saloon lifejacket locker, which most passengers could not reach unaided, did not contribute but increased risk (Safety Issue). The master's split-duty roster did not contribute. Tidal shortfall and steering or propulsion failure were ruled out.

What has been done as a result

The Harbour Master, Kanooka Heads, funded the mangrove clearing. The sightline was restored on 24/07/2026 and a three-yearly trimming cycle programmed. Local notice LNM-2026-14 advised the beacon's 3 M range, and Tarrakin Ferries reissued its passage plan appendix with the corrected

figure. The investigation recommended a night passage plan with a hold point, and a sightline test in place of the 50 m vegetation scope in the services agreement.

Safety message

A passage plan that names a primary means of position fixing should say what to do when it fails, and stopping on flat water costs nothing. Moving a route into darkness changes its risk, not only its timetable. Authorities responsible for aids to navigation should confirm that a light can be seen from the water it serves, not only that it is lit.

The occurrence

Before the voyage and the departure

On 07/07/2026, MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers, was operating Tarrakin Ferries' service between Muntari Island and the town wharf at Kanooka Heads.

The morning master worked from 05:40 to 13:05. The chart plotter's event log recorded display freezes at 10:42 and 11:17 local, each followed by an automatic watchdog reset. The outgoing master stated that both occurred outside the Weerona Channel and that each time the display recovered by itself after a minute or two.

At about 13:05 the masters changed over at the town wharf. Their accounts agreed that the outgoing master told the relieving master about the two freezes, and that the relieving master replied the plotter had done the same to him the month before. Neither entered the fault in the wheelhouse defect book, which held no entry of any kind for 07/07/2026 before the occurrence. The afternoon's two round trips were described as normal.

TF18, the last service of the day, was one of the winter timetable's evening returns, made in darkness since 01/06/2026. It departed Muntari Island at 17:35:02 with 38 passengers, a master, a deckhand and a kiosk attendant on board, and the master reported the departure to Port Control on VHF channel 12. Sunset had been at 17:02 and civil twilight ended at 17:29, so the channel transit was made in full darkness. The master described clear conditions, a light westerly and no sea in the entrance.

The channel transit

The vessel entered the Weerona Channel from the north abeam the No. 1 beacon at 18:03:58, 2 m east of the charted centreline at 8.4 kt, on a course over the ground of 195 degrees. At 18:04:10 the deckhand asked whether to stay on the bridge, and at 18:04:14 the master sent her to collect fares. She stated she was in the saloon from about 18:05 and could not speak to anything on the bridge after that.

At 18:05:30 the vessel was abeam the No. 2 beacon, 1 m west of the centreline and 0.90 nm north of the No. 4 beacon. The bridge audio recorded the master saying "Two ... good."

At 18:07:06, with the vessel 3 m west of the centreline and 0.68 nm north of the No. 4 beacon, the plotter event log recorded that the display process had stopped responding. The cause code matched the morning events, but no reset was recorded and the unit did not recover. The chart picture stayed on screen with the vessel symbol stationary, and the display did not accept touch input. The bridge audio recorded the master saying "Oh, come on" at 18:07:11, sounds consistent with repeated screen taps at 18:07:26, and the words "frozen again", partly unintelligible, at 18:07:44.

The master continued at the same speed and looked out for the No. 4 starboard-hand beacon on the western side of the channel. He told the investigation he did not consider stopping, because in his mind the visual transit was the backup.

From 18:07:06 the heading increased steadily from 195 to 200 degrees over about a minute and was then held. The rate of turn peaked at 2.1 degrees per minute at 18:07:40 and returned to zero by 18:08:50. Over this period both engines ran at 1,780 and 1,784 rpm with no alarm and no throttle movement, steering hydraulic pressure stayed within its normal band, and rudder angle followed

helm demand within 1.5 degrees. The master later stated that he had eased the vessel five degrees to starboard while searching for the beacon, without deciding to do so.

At 18:08:31, 12 m west of the centreline and 0.47 nm north of the beacon, the master was recorded saying "Right. Where's four." The vessel reached the western edge of the dredged channel at about 18:09, at 8.0 kt on a heading of 199 degrees, and from there set west at an average of 12 m per minute.

At 18:09:40 the audio recorded a sound consistent with a window or door being slid open. The master stated he had opened the wheelhouse window in case the glass was the problem. He described the beacon as hard to pick out against the lights of the town and the new car park behind it, but twice told the investigation he could not say whether that lighting was why he could not see it.

A range and visibility check on 18/07/2026 found that the beacon's light was not visible from the channel between 0.84 nm and 0.34 nm north of it, where the sightline passed through mangrove canopy on Weerona Islet.

At 18:10:02, 39 m west of the centreline and 0.25 nm north of the beacon, the master was recorded saying "There's four. Keep her over." He recalled that the beacon was then dead ahead rather than on the starboard bow, and that he understood at once he was on the bank side. By 18:11:00 the vessel was over Boolarong Bank, 58 m west of the centreline. Speed over the ground had fallen from 8.4 kt to 7.6 kt since 18:07 with no throttle movement.

The grounding

The echo sounder shallow alarm, set at 0.9 m under the keel since commissioning in 2019, sounded at 18:11:48 with the vessel 70 m west of the centreline at 7.1 kt. The master was recorded saying "No no no" at 18:11:50, and at 18:11:52 both throttles moved together to the astern detent.

At 18:12:04 the vessel made contact with Boolarong Bank at 6.8 kt over the ground, with astern thrust developing but not yet at full. It covered 58 m in the 16 seconds between the alarm and contact, against a trials crash-stop distance from 8 kt of 92 m. The audio recorded a heavy impact followed by grinding and vibration for approximately 9 seconds.

The vessel stopped within 11 m and came to rest heeled 4 degrees to starboard on a falling tide, 74 m west of the charted centreline, 54 m outside the dredged channel and 34 m west of the No. 4 beacon. The bed at the contact point was 0.50 m below chart datum. Furrows consistent with the starboard hull and skeg ran 11 m from the point of first contact on a heading of 197 degrees.

The response

The master stopped both engines at 18:12:31 and did not restart them. At 18:12:40 he told passengers over the public address that the vessel had touched the bottom and was stopped and safe.

At 18:13:04 he broadcast a pan pan to Kanooka Heads Port Control on VHF channel 16, giving the vessel's position and reporting 38 passengers and three crew on board, no injuries reported, no pollution, the vessel stable and upright, and that he was not attempting to refloat. He requested the workboat for a passenger transfer, and Port Control acknowledged at 18:13:29 and called it out. The master later explained he chose not to refloat because the tide was falling, the bottom was unknown and 38 people were aboard.

At 18:14:52 the master asked passengers to put on lifejackets. At 18:16:20 the deckhand reported water in the No. 2 void after sounding it at his direction. At 18:17:05 the master reported to Port

Control slow ingress into the No. 2 starboard void, approximately 75 mm on the sounding rod, being monitored every ten minutes.

The deckhand handed out lifejackets working from aft forward. In the forward saloon the locker was behind the kiosk servery counter, where three passengers were standing. She had to ask them to move, pass through a narrow gap at the end of the counter, and hand the jackets over it. At 18:22:40 she reported that everyone was wearing a lifejacket.

The workboat came alongside the port quarter at 18:41. At 19:26:08 the master reported all 38 passengers ashore and the three crew remaining on board. One passenger reported a grazed shin and two were treated at the wharf for distress. There were no serious injuries.

The vessel was refloated on the rising tide at 00:20 on 08/07/2026 and towed to the town wharf, by which time the No. 2 void held approximately 260 mm of water. The damage was:

- starboard hull bottom plating set up over 4.1 m, with a 180 mm split
- starboard propeller blades bent, and the shaft bent 1.4 mm
- starboard rudder stock distorted 3 degrees
- starboard skeg torn away.

The port hull, propulsion and steering were undamaged. The vessel was withdrawn from service, and the Weerona Channel was closed to commercial traffic for 19 hours.

Context

Personnel information

Master

The master held a certificate of competency, master less than 35 m, issued in 2012. He had 14 years' experience as a master, about 19 years in commercial marine operations and 11 years with Tarrakin Ferries, all on the Muntari Island route, with about 4,100 transits of the Weerona Channel. His most recent route check, on 14/02/2026, was satisfactory, and he had no prior occurrence history. He commanded 26 of TF18's 31 darkness transits of the channel between 01/06/2026 and the occurrence.

Outgoing master

The outgoing master held the same certificate. He had 9 years as a master, about 15 years in commercial marine operations, 6 years with the operator and 4 on the route, with about 1,400 channel transits. His most recent route check, on 09/11/2025, was satisfactory. He made the other 5 darkness transits. On 07/07/2026 he worked from 05:40 to 13:05, on day six of a twelve-day tour.

Deckhand

The deckhand had 7 months with the operator and about 3 years in commercial marine operations. She held a certificate of safety training and a coxswain grade 2 qualification, was in her first winter season on the route, and completed vessel familiarisation and drills in 12/2025.

Duty and rest

The route was worked on 12 days on and 2 days off, and 07/07/2026 was the master's ninth consecutive day. His duty ran 05:40 to 08:40 and 13:10 to 19:20, giving 9 hours 10 minutes of duty across a 13 hours 40 minutes spread, with an unpaid 4 hours 30 minutes break at home. Both were inside the operator's limits of 10 hours of duty and 14 hours of spread, and inside the state crewing determination. The master recorded 6 hours 15 minutes of sleep on the night of 06/07/2026, in a fatigue diary the operator asks masters to keep but does not collect.

Vessel information

General

MV Tarrakin Star was a 24.4 m aluminium catamaran passenger ferry (beam 8.2 m, static draught 1.35 m), certificated for 96 passengers and operated as a class 1D vessel on this route. Its annual survey on 11/03/2026 left no outstanding conditions, and it was last slipped in 03/2025.

The wheelhouse defect book held eleven entries in the 90 days before the occurrence, none concerning the chart plotter, GPS antenna, echo sounder, steering gear, engines or propulsion controls. The maintenance system held no work order, fault report or correspondence about the chart plotter in 2026.

Chart plotter

The chart plotter was a 12-inch multifunction display fed by a separate GPS antenna unit, running chart software revision 4.11 on firmware issued in 03/2024. It was not a statutory item for this vessel and route, but the passage plan named it the primary means of position fixing.

On 21/07/2026 an approved electronics contractor bench tested it for 46 hours with the same antenna and cabling. The fault did not recur and no defect was found. The contractor reported that

the manufacturer knew of cause code 0x51 and had addressed it in firmware issued in 11/2024, not applied to this unit. The operator did not hold the bulletin and had no subscription to the manufacturer's bulletins.

The outgoing master believed the plotter had frozen three or four times in the past year, to him and other masters, and stated that none had been written up.

Echo sounder

The shallow alarm was set to 0.9 m under the keel, the commissioning value from 2019, never changed. The vessel's trials crash stop distance from 8 kt was 92 m.

Steering and propulsion

On 12/07/2026 both steering rams, the tiller arms, the hydraulic system and the autopilot were removed and function tested. All were serviceable, with no internal leakage, no air in the system and no sign of a pre-existing defect, and the system passed a full lock-to-lock timing test inside the specified period.

Lifejacket stowage

The vessel carried 120 adult and 20 child lifejackets in four lockers. The forward saloon locker held 40 adult lifejackets in the bulkhead behind the kiosk server counter. The arrangement approved at build in 2011 had an open bench there, leaving the locker face clear. The counter was fitted in a 2019 refit under a minor modification notice that did not trigger re-survey of the safety equipment arrangement.

A survey on 16/07/2026 found the locker face could not be reached from 22 of the 34 forward saloon seats without passing behind the counter through a 0.6 m gap at its inboard end. The muster placard directed passengers to a locker not visible from the seats. The same counter was fitted to the operator's two other 24 m vessels.

The Weerona Channel and its aids to navigation

The channel

The Weerona Channel was a dredged recreational and commercial access channel, 40 m wide and at least 3.2 m below datum, with Boolarong Bank, which dried at low water, to the west. The route risk assessment recorded a tidal range of 1.4 m and a single unlit reach between the No. 2 and No. 4 beacons. Kanooka Shire Council managed the waterway through the Kanooka Heads Port Authority, and six beacons marked the channel.

The No. 4 beacon

The No. 4 beacon showed Fl G 2.5s from a lantern 4.2 m above chart datum. In 11/2025 the Council's waterways energy programme converted it from mains to a solar lantern of 3 nautical miles nominal range, against 6 for its predecessor. The hydrographic service was notified on 04/02/2026, but the chart correction was still pending on 07/07/2026. No local notice to mariners was issued, no operator was told, and the operator's passage plan still recorded 6 M. A measurement from the port workboat on 18/07/2026 gave a luminous range of 2.6 nm.

Weerona Islet lies 240 m north-west of the beacon. Its mangrove canopy was 2.1 m in the 2019 aerial survey and 5.8 m on 18/07/2026. Work order WO-118 (09/2023) recorded regrowth screening the beacon from the northern approach between about half a mile and a quarter mile, and recommended trimming to 2 m on a three-yearly cycle at an estimated \$18,500. It was deferred as unfunded at the 2023, 2024 and 2025 works rounds. WO-176 (03/2026) recorded the northern

sightline as fully closed. An Authority file note recorded that the islet was outside the 50 m vegetation scope of the services agreement, so the work competed with boat ramps and pontoons for general waterways funding rather than with aids to navigation.

The Authority's monthly check was made from the town wharf, 2.1 nm south of the beacon, where the islet is not in the line of sight. Every monthly check from 01/2024 to 07/2026, the last on 02/07/2026, recorded the beacon as operating. No defect or notice had ever been recorded against it, the Authority held no record of operator enquiries about the channel's aids, and it had never provided its register to Tarrakin Ferries.

Reporting of aids by users

Both masters interviewed had noticed the No. 4 beacon becoming harder to see over about two years, and neither reported it. The master told the investigation: "There's no form for the beacon looks dim." The operator's reporting system had a navigation hazard category, unused on this route in three years, which no master interviewed knew existed.

Foreshore lighting

The Kanooka Heads foreshore redevelopment, commissioned on 22/05/2026, included 14 LED floodlights on 12 m columns lighting a boat ramp and car park 300 m east of the town wharf. None was shielded to seaward, and each presented about 4,800 candela toward the channel. From 0.5 nm north of the No. 4 beacon on the inbound track, the continuously lit array lay behind the beacon, within 4 degrees of its bearing and 1.5 degrees of its elevation, while the beacon flashed green for 0.4 seconds every 2.5 seconds. The Council's planning unit approved the redevelopment without consulting the Port Authority's aids to navigation officer, and no assessment of the lighting against the channel's aids was carried out or required.

Regulatory framework

The Waterways Aids to Navigation Code WAN-2 imposes its stronger duties only on declared shipping channels. Section 4.2 provides that there the responsible authority "must record its availability, must achieve an availability of not less than 99.0 per cent measured annually, and must notify a change to the aid's character, position or range by notice to mariners before the change takes effect." Section 4.3 provides that elsewhere the authority "must be able to demonstrate that the aid is inspected at intervals it has determined and recorded." The Weerona Channel is not a declared shipping channel, so section 4.2 did not apply to its six beacons, and the Authority was not in breach of the code.

Schedule 2 of the services agreement between the Council and the Port Authority provided:

- inspection of each lit aid "by daylight once per quarter", with confirmation of its operation "by observation from the town wharf once per month" [S2.1]
- vegetation beyond 50 m of an aid "is not within the scope of this agreement" [S2.4]
- "There is no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid." [S2.7]

Operator procedures

Passage planning

Operations Manual OPS-3 revision 6 (02/2023) required the transit to be "made on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified visually before the vessel passes the No. 1 beacon" [3.4]. Section 3.6 required reduced speed, and

consideration of anchoring or returning to open water, in reduced visibility only. OPS-3 made no provision either way for a second person on the bridge during a channel transit.

Appendix A, issue 2, the current passage plan, specified 8 knots and an inbound course of 195 degrees. It named the chart plotter as primary means of position fixing and the "visual transit of the No. 4 beacon and the town wharf leading lights" as secondary, and noted that the plan "is written for the daylight transit. Night transits are not scheduled on this route." There was no night passage plan, no guidance for loss of the plotter, and no abort or hold point. The leading lights lay 2.1 nm beyond the No. 6 beacon and did not resolve as two lights until the vessel was inside the channel's southern third.

Route risk assessment and the winter timetable

Route Risk Assessment RRA-07, issue 2 (03/2023), was current, with no issue 3. It stated: "daylight transits only. A night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan."

Change record CHG-2026-014 moved TF18 from 16:15 to 17:35 from 01/06/2026, assessing it as "timetable change only", with "No new hazard identified" and "Route risk assessment review required: No." The Operations Manager approved it on 22/04/2026. It did not refer to the daylight-only restriction. RRA-07 was not reopened, no night passage plan was written, and the General Manager Operations' approval was neither sought nor given.

Defect reporting

Operations Manual OPS-9 revision 4 (08/2024) required that:

- a defect affecting seaworthiness or a statutory item be reported immediately [9.1]
- all other defects be entered "at the end of the shift in which they are found" and reviewed by the Operations Manager weekly [9.2]
- at a change of master, the outgoing master brief the relieving master on "the state of the vessel, any defect, and anything else affecting the conduct of the service" [9.3].

Nothing required the handover brief to be written down, or the relieving master to confirm that a defect mentioned in it had been entered in the defect book.

Meteorological, tidal and light conditions

At 18:00 the Kanooka Heads automatic weather station recorded wind 305 degrees at 8 kt, visibility greater than 10 km, few cloud at 3,500 ft and no rainfall. At 18:45 the port workboat reported a slight sea in the channel with wind ripple only.

Predicted high water was 1.66 m at 15:20 and low water 0.30 m at 21:34. The predicted height for 18:12 was 0.94 m, and the Port Authority gauge recorded 0.91 m then, with a largest departure from prediction of 0.07 m in the 12 hours to 20:00. Sunset was at 17:02 and civil twilight ended at 17:29. The moon was a waning crescent, 2% illuminated, which had risen at 06:12.

Recorded data

The investigation used:

- the Port Control AIS receiver log and the vessel's engine monitoring and steering recorder readout, issued 15/07/2026
- the plotter's internal event log, recovered on 09/07/2026
- the echo sounder alarm log, recovered on 09/07/2026

- a transcript of VHF channels 12 and 16 and the bridge camera audio, made on 16/07/2026 and verified by the Port Authority's traffic officer.

These sources were logged in UTC. All times in this report are local (AEST, UTC+10).

Safety analysis

External Influences

Aids to navigation on this waterway carry no availability standard and no duty to tell users anything

The obligations on the authority responsible for the Weerona Channel's beacons are set by the Waterways Aids to Navigation Code WAN-2. Section 4.1 requires that an aid be maintained "so that it performs the function for which it was established." The specific duties sit in section 4.2, which requires the responsible authority to record an aid's availability, to "achieve an availability of not less than 99.0 per cent measured annually," and to "notify a change to the aid's character, position or range by notice to mariners before the change takes effect." Those duties apply only where an aid is established on a declared shipping channel. The Weerona Channel is not a declared shipping channel, so section 4.3 applies instead. It requires only that the responsible authority "be able to demonstrate that the aid is inspected at intervals it has determined and recorded." None of the six beacons in the channel therefore carried an availability target, a duty to record availability, or a duty to tell the operators using the channel about a change to character or range. The code's text and the channel's status are documentary and not in dispute, and it is virtually certain that the framework applied as described. The Kanooka Heads Port Authority was not in breach of the code.

The services agreement between Kanooka Shire Council and the Authority set the inspection interval that section 4.3 left to the Authority. Under schedule 2, each lit aid is inspected "by daylight once per quarter" and its operation is confirmed "by observation from the town wharf once per month." Vegetation "beyond 50 m is not within the scope of this agreement," and there is "no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid." The town wharf is 2.1 nm south of the No. 4 beacon, on its southern side. Nothing in the agreement requires an aid to be observed from the approach a vessel actually uses. The Authority was not in breach of the agreement either.

The two organisational factors that bore on the No. 4 beacon both arose inside this framework. When the beacon was converted to solar in 11/2025, its nominal range fell from 6 to 3 nautical miles. The Authority lodged a notification with the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator was told. On the night, Tarrakin Ferries' passage plan appendix still recorded the beacon at 6 M. The mangrove screening on Weerona Islet, 240 m north-west of the beacon, was identified in work order WO-118 in 09/2023 and deferred as unfunded at the 2023, 2024 and 2025 works rounds. The Authority's file note against WO-176 recorded that the islet lay outside the 50 m scope of schedule 2.4, so the work competed with boat ramps and pontoons in the general waterways vote. Meanwhile the monthly check recorded the beacon as observed and operating in every month from 01/2024 to 07/2026, from a position at which the islet is not in the line of sight. No defect was ever recorded against the beacon, no notice about it was ever issued, and the aids to navigation register was never provided to the operator. The argument that each of these factors contributed to the beacon not being visible from the channel is set out under Organisational Influences.

The case for this framework having influenced the occurrence is that every mechanism in the code that would have carried the beacon's condition to the people using it stops at the boundary of a declared shipping channel. That includes a recorded availability figure and a notice to mariners. Under section 4.2, the halving of the nominal range would have required a notice to mariners before it took effect. That duty is the one mechanism identified that would have put the range

change in front of the operator, whose passage plan still carried the earlier figure on the night of the occurrence.

There is evidence against that view, and it is substantial. Nothing in the code prevented the Authority from notifying operators, trimming the mangroves or measuring the light. The evidence indicated that these things were not done for reasons of funding and priority, not because the code permitted their omission. In addition, an availability target of 99.0 per cent would not have detected the problem at all. The lantern was lit and working throughout, and the difficulty was that it could not be seen from the inbound track, which no availability figure measures. On that reading, the scope of the code was background to the occurrence rather than a cause of it.

On balance, the investigation found that the absence of an availability standard and a notification duty likely contributed to the occurrence. It did so by allowing the reduction in the beacon's nominal range and the deferral of its sightline clearing to proceed without any requirement that the operator be told. The availability argument against influence was accepted as far as it goes. A notification duty under section 4.2 would have covered the change in range but not the vegetation screening, and an availability target would have covered neither. The finding therefore rests mainly on the notification pathway for the range change. It is held at a lower level of confidence than the other contributing factors in this analysis, and it should be read as the investigation's weighed conclusion on finely balanced evidence rather than as a settled matter.

Organisational Influences

Winter timetable moved the last service into darkness without reopening the route risk assessment

The current route risk assessment for the Weerona Channel, RRA-07 issue 2 (03/2023), names the single unlit reach between the No. 2 and No. 4 beacons as a hazard. It also carries an explicit operating restriction: "daylight transits only. A night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan."

Change record CHG-2026-014 moved service TF18 from 16:15 to 17:35 for the winter timetable from 01/06/2026. It assessed the change as a "timetable change only", recorded "No new hazard identified", and answered "No" against route risk assessment review. The Operations Manager approved it on 22/04/2026, one level below the person the restriction names.

The change record does not refer to the daylight restriction. RRA-07 was not reopened and no issue 3 exists. No night passage plan was written, and the General Manager Operations' approval was neither sought nor given. From 01/06/2026 every TF18 transit of the channel was made after the end of civil twilight, and TF18 ran 31 times in darkness before the occurrence. Both documents are current controlled copies and neither is in dispute. It is therefore virtually certain that the route moved into darkness without the risk assessment being reopened.

The change process asked whether the change introduced anything new. It did not ask whether the existing assessment still held under the changed conditions. A review of RRA-07 against a night operation would have reached the unlit reach the assessment already identified. The restriction itself would then have required a specific passage plan before the first night service ran.

It was likely that this contributed to the occurrence. It was not assessed as more certain than that, for two reasons. A review might have produced a plan that said no more than the daylight plan does. In addition, the Operations Manager's understanding of what the restriction required was given only verbally, so the reasons the change was assessed in this way could not be fully determined. The absence of a night passage plan and of any hold point on loss of the primary means of fixing is analysed under Risk Controls.

Beacon converted to solar, nominal range halved, and no operator was told

In 11/2025 the No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern is rated at a nominal 3 nautical miles, against 6 for the lantern it replaced. The character, Fl G 2.5s, did not change.

A notification was lodged with the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued, and no operator using the channel was told. A measurement from the port workboat on 18/07/2026 gave a luminous range of 2.6 nm. The operator's current passage plan appendix still listed the aid as "No. 4 (Fl G 2.5s, range 6 M)".

The conversion, the lantern rating, the lodgement and the absence of any local notice are all documentary, and the range was measured independently. It is virtually certain that the change was made and that no operator was told of it. As a result, the master was navigating to a written plan that described a light more than twice the range of the one actually fitted.

A weaker light is harder to acquire. The master began looking for the No. 4 beacon at 18:08:31 and called it at 18:10:02, 91 seconds later. It was likely that the reduced range contributed to the occurrence. That assessment is moderated because most of the search took place inside the arc where the light is screened entirely, and a screened light is not made visible by being brighter. Whether new foreshore floodlighting behind the beacon also affected its detection could not be resolved, and that question is discussed separately.

Clearing of the beacon's sightline deferred unfunded for three works rounds

Work order WO-118, raised in 09/2023, recorded that mangrove regrowth on Weerona Islet screened the No. 4 beacon from the northern approach between about half a mile and a quarter mile. It recommended trimming to 2 m on a three-yearly cycle, at an estimated cost of \$18,500. The work was deferred as unfunded at the 2023, 2024 and 2025 works rounds, and each time it was referred to the next round. Work order WO-176, raised in 03/2026, recorded the same sightline as fully closed on the northern approach.

The Port Authority's file note against WO-176 recorded why the work had not been funded. The islet is 240 m from the beacon, which places it outside the 50 m scope of schedule 2.4 of the services agreement. The work therefore competed with boat ramps and pontoons in the general waterways vote, rather than being funded as aid to navigation work.

The canopy was given as 2.1 m in the Council's 2019 aerial survey. It measured 5.8 m on 18/07/2026, against a lantern height of 4.2 m above chart datum.

The Authority's monthly check recorded the beacon as observed and operating in every month from 01/2024 to 07/2026. That check was made from the town wharf, 2.1 nm south of the beacon, a position from which the islet is not in the line of sight. No defect was ever recorded against the beacon, and no notice about it was ever issued to a waterway user. The aids to navigation register was never provided to Tarrakin Ferries.

These matters are documented in the Authority's own register and were confirmed by measurement on site. It is virtually certain that the clearing was deferred as described. The 18/07/2026 check found the light not visible from the channel between 0.84 nm and 0.34 nm north of the beacon. The plotter froze at 0.68 nm, and the vessel crossed the channel's western edge at about 0.40 nm. The whole of the period in which stopping in the channel remained available therefore fell within the screened arc.

It was very likely that the deferral contributed to the occurrence. The remaining doubt allows for the possibility that the master might not have picked up the light in that time even had it been

visible. The framework that left this aid without an availability standard or a duty to notify users was analysed under External Influences.

Risk Controls

No night passage plan for the channel, and no hold point if the primary means of fixing is lost

It is virtually certain that no night passage plan existed for the Weerona Channel. The operator confirmed this. OPS-3 3.4 required the transit to be "made on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified visually before the vessel passes the No. 1 beacon." Appendix A, issue 2 (02/2023), the current issue, named the chart plotter as the primary means of position fixing. It named the "visual transit of the No. 4 beacon and the town wharf leading lights" as the secondary means. Its only reference to darkness was its closing note: "Night transits are not scheduled on this route."

The plan said nothing about losing the plotter during the transit. It named no abort criterion and no hold point. The secondary means was also weaker than it appeared. The town wharf leading lights lie 2.1 nm beyond the No. 6 beacon and do not resolve as two lights until a vessel is inside the channel's southern third. When the plotter froze, 0.68 nm north of the No. 4 beacon, that beacon was therefore the whole of the secondary means. OPS-3 3.6 required speed to be reduced and anchoring to be considered, but only "in reduced visibility". Visibility was not reduced on the night, so 3.6 did not apply.

The organisational origin of this gap, a winter timetable change that moved the last service into darkness without reopening the route risk assessment, is discussed under Organisational Influences. The decision to continue the transit at eight knots after the plotter froze is discussed under Individual Actions.

It is very likely that the absence of a hold point contributed to the transit continuing. The display froze at 18:07:06 and the vessel crossed the channel's western edge at about 18:09. Stopping was therefore available for about 114 seconds, on flat water with no other traffic. A named hold point would have applied throughout that time. Both masters interviewed said independently that stopping was the right response, and both said that nothing told them to do so. The investigation did not grade the influence higher than very likely for one reason. A master who did not think of stopping unprompted might not have applied a written rule under the same pressure.

Change-of-master defect brief is verbal, with nothing checking that what is said gets written down

It is virtually certain that the defect reporting arrangements worked as both masters described them. OPS-9 9.1 required immediate reporting only of a defect affecting seaworthiness or a statutory item. The chart plotter was not a statutory item on a class 1D vessel on this route, even though the passage plan named it as the primary means of position fixing. All other defects fell under 9.2, which required them to be "entered in the vessel defect book at the end of the shift in which they are found". For a fault that comes and goes, that is no trigger at all.

Under 9.3, the outgoing master was to brief the relieving master on "any defect". Nothing required that brief to be written down, and nothing required the relieving master to confirm that a defect had been booked. Both masters described the same understanding independently. The book was for what was broken at handover, and anything that had come good was spoken about.

On the day, the plotter's event log showed freezes at 10:42 and 11:17, and the unit recovered itself both times. At the change of master at about 13:05, the outgoing master passed this on verbally,

and no entry was made. The outgoing master believed the plotter had frozen three or four times on the vessel in the past year, and that none of those freezes had been written up. The maintenance system held no plotter entry of any kind for 2026. The operator did not hold the manufacturer's firmware bulletin issued in 11/2024, which addressed cause code 0x51, and the fix had not been applied. The individual action at the handover is discussed under Individual Actions. This analysis explains why the omission occurred. It does not excuse it.

The evidence supported two readings of influence:

- **Against influence.** An entry made at 13:05 would have been reviewed by the Operations Manager at the end of that week, not that afternoon. On this reading, the vessel would have sailed TF18 either way. A different outcome on the night would have required a same-day decision that nobody in the system was obliged to make. The handover rule would then be a genuine finding about the fleet, but not a cause of this grounding.
- **For influence.** The significance lies less in the single entry not made that day than in the pattern over the year. Three or four freezes on the primary means of position fixing never reached anyone ashore who could act on them. As a result, the firmware fix was never applied, and the master met the fault on the night with no reason to treat it as more than a nuisance.

The investigation found the second reading likely, while accepting that the first has force.

Forward saloon lifejacket locker not reachable from most of the saloon it serves

It is virtually certain that the forward saloon locker could not be reached from most of the saloon it served. The locker held 40 adult lifejackets and was mounted behind the kiosk servery counter. The investigation's survey on 16/07/2026 found that the locker face could not be reached from 22 of the 34 forward saloon seats without passing behind the counter through a 0.6 m gap at its inboard end. With three people standing at the counter, as there were on the night, that gap was the only route. The muster placard directed passengers to a locker they could not see from their seats.

The deckhand reported at 18:22:40: "There's two in the forward saloon I had to go round the servery for, the locker's behind the counter and there's people standing at it."

It is unlikely that the locker's position had any bearing on the grounding or its outcome. All 38 passengers were in lifejackets within about eight minutes and were ashore within 74 minutes. The vessel was upright and stable throughout.

The arrangement was very likely important nonetheless. The muster on the night was unhurried and took a couple of minutes longer than it should have. In a fire, a capsized risk, a night evacuation into the water, or with a deckhand injured in the impact, a locker that passengers cannot reach without crew help would separate a saloon that equips itself from one that waits.

The underlying gap is in the modification route. The design approved at build in 2011 had an open bench in that position, with the locker face clear. The counter was fitted during a 2019 interior refit under a minor modification notice, and that notice did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels.

Local Conditions

Chart plotter display froze during the channel transit and did not recover

It is virtually certain that the chart plotter display froze at 18:07:06 and did not recover. Three independent sources agree on this, and none contradicts the others:

- The unit's internal event log recorded the display process as not responding at that time, with no accompanying reset.
- The bridge audio recorded the master saying "Oh, come on" at 18:07:11, then the sound of repeated screen taps at 18:07:26, then "(unintelligible) frozen again" at 18:07:44.
- The master stated that the chart picture stayed on the screen with the vessel symbol stationary, and that the display would not accept touch input.

An approved contractor bench tested the unit for 46 hours on 21/07/2026 with the same antenna and cabling. The fault did not recur and no defect was found. That result bears on why the display froze, not on whether it froze. The event log recorded the same cause code, 0x51, at 10:42 and 11:17 that morning, when the unit recovered itself. The contractor identified that code as known to the manufacturer and addressed by firmware issued in 11/2024, which had not been applied to this unit. The investigation could not establish why the unit failed to recover on this occasion. How the morning freezes were passed on at the change of master is discussed under Individual Actions.

It is extremely likely that the freeze influenced the occurrence. The vessel held the centreline while the plotter worked. It was 1 m west of the centreline abeam the No. 2 beacon and 3 m west when the display froze. Within two minutes it had reached the channel's western edge. The master began searching for the No. 4 beacon 85 seconds after the freeze. From that point he was navigating without the primary means of position fixing named in the passage plan. The decision to continue at about 8 kt is discussed under Individual Actions.

No. 4 beacon not visible from the channel over the half mile before the vessel left it

It is extremely likely that the No. 4 beacon could not be seen from the channel between 0.84 nm and 0.34 nm north of it. This was measured on 18/07/2026 from the port workboat, on the inbound track and at the same time of night. Over that stretch, the sightline passed through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. The canopy was measured at 5.8 m, against a lantern height of 4.2 m above chart datum. The measurement was made eleven days after the occurrence, and the canopy was not surveyed on the night itself. For that reason the finding is put at extremely likely rather than higher.

The bridge audio is consistent with the measurement. The master said "Right. Where's four" at 18:08:31, when the vessel was 0.47 nm north of the beacon. A window or door was heard sliding open at 18:09:40. He said "There's four. Keep her over" at 18:10:02, at 0.25 nm. His whole 91-second search took place inside the screened arc. Both masters interviewed raised the beacon without being prompted. Neither had ever reported it.

It is very likely that the screening influenced the occurrence. With the plotter frozen, the beacon was the only means of fixing available to the master. It was hidden for the whole period in which the vessel set west, and the vessel crossed the channel's western edge during the search. The qualification is that the vessel was already 12 m west of the centreline when the master began looking. The screening therefore cannot account for the start of the departure. The solar conversion and the deferred sightline clearing that produced this condition are discussed under Organisational Influences.

New foreshore floodlighting behind the beacon reduced the master's ability to pick it up

The investigation examined whether new foreshore floodlighting made the beacon harder to detect, and could not settle the question.

Several things supported the possibility:

- The master described looking "into a wall of light" and called the beacon "a nothing" against the town and the new car park.
- The outgoing master, independently, described it as "washed out altogether from up the top".
- The 18/07/2026 survey found 14 unshielded LED floodlights on 12 m columns at the new boat ramp, each presenting about 4,800 candela toward the channel. They were commissioned on 22/05/2026.
- From 0.5 nm north of the beacon, the array lies behind it within 4 degrees of the same bearing and within 1.5 degrees of the same elevation. The floodlights are lit continuously, while the beacon shows a green flash of 0.4 seconds every 2.5 seconds.
- The Port Authority's aids to navigation officer was not consulted on the development, and no assessment of the lighting against the channel's aids was required or made.

Several things weighed against it:

- Over the stretch where the master was looking, the mangrove canopy screened the beacon. It would not have been visible against a dark background either.
- The one point where the alignment was measured, 0.5 nm north, lies inside the screened arc.
- The master picked the light up once it cleared the islet, with the floodlights in the same place and no dimmer.

No measurement exists of what could be seen on the night. Nothing establishes a detection threshold for a 0.4-second green flash against that background. While the mangroves screen the arc, any separate effect of the floodlights may not be separable. The matter remains unresolved, and no finding is drawn from it.

Water level below prediction reduced the depth over Boolarong Bank

The possibility that the water level was below prediction was ruled out as very unlikely. At 18:12 the Port Authority's gauge read 0.91 m against a predicted 0.94 m, and the vessel touched because it was 54 m outside the dredged channel.

Day nine of twelve, on a split duty across a thirteen hour forty spread

It is virtually certain that the master was on his ninth consecutive day of a 12-day tour. His split duty ran from 05:40 to 08:40 and from 13:10 to 19:20, giving 9 hours 10 minutes of duty across a spread of 13 hours 40 minutes. Both figures were inside the operator's limits and the state crewing determination. He recorded 6 hours 15 minutes of sleep the night before.

It is unlikely that this influenced the occurrence. The recorded data showed no sign of fatigue:

- The centreline was held closely in the channel.
- The master responded to the shallow alarm in 4 seconds, against a fleet figure of 6 to 9 seconds.
- His radio work after the grounding was complete and correct at the first attempt.
- He decided not to refloat in about thirty seconds.

The one decision fatigue could have degraded was not considering stopping. Both masters said, independently, that no document prompts that consideration. The outgoing master, who was not tired, described the same understanding.

It is likely that this exposure is nonetheless important. It is a standing feature of the winter roster:

- The last darkness transit falls at the end of a spread within twenty minutes of the operator's limit, on the ninth to twelfth consecutive day.

- The operator asks masters to keep a fatigue diary but does not collect it.
- No relief master is rostered.

Individual Actions

Plotter fault passed on verbally at the change of master and not entered in the defect book

It is virtually certain that the plotter fault was passed on verbally at the change of master and not entered in the defect book. The plotter's event log showed two display freezes during the morning shift, at 10:42 and 11:17, and the unit recovered from both by itself. Both masters gave the same account of the handover at about 13:05 at the town wharf. The outgoing master told the relieving master about both freezes, and the relieving master replied that the unit had done the same to him the month before. Neither entered the fault in the wheelhouse defect book or reported it to the Operations Manager. The defect book held no entry of any kind for 07/07/2026 before the occurrence. Both masters volunteered the omission before they were asked about it.

The omission was consistent with what OPS-9 required of them. The plotter was not a statutory item, so the fault fell under 9.2, which states that such defects "shall be entered in the vessel defect book at the end of the shift in which they are found, and will be reviewed by the Operations Manager weekly." For an intermittent fault that clears itself, that gave no clear trigger. Nothing required the brief under 9.3 to be written down, and nothing required the relieving master to confirm that a defect mentioned in it had been entered. The matter is therefore one of practice, and the conditions that allowed it are discussed under Risk Controls, where the verbal change-of-master brief is addressed.

The maintenance system held no work order, fault report or correspondence about the plotter in 2026, and the operator had no subscription to the manufacturer's bulletins. Cause code 0x51, which the unit logged on both morning freezes, was addressed by firmware issued in 11/2024 that had never been applied. An entry would have placed a repeating fault in front of the people who could obtain that fix or withdraw the vessel. It was likely that the omission contributed to the plotter freezing in the channel. The likelihood was not higher because a single entry made at 13:05 and reviewed on a weekly cycle would not, by itself, have stopped TF18 sailing that evening.

Transit continued at eight knots after the plotter froze, rather than stopping in the channel

It is virtually certain that the transit continued at about eight knots after the plotter froze. The AIS track, the engine recorder, the bridge audio and the master's own account agree. Speed over the ground held at 8.4 kt to 18:08. The later fall to 7.6 kt by 18:11 occurred with no throttle movement and was consistent with the drag of shoaling water. The audio captured "(unintelligible) frozen again" at 18:07:44 and "Right. Where's four" at 18:08:31, 85 seconds after the freeze, with nothing suggesting that stopping was considered. The master stated that he did not consider stopping, that he went straight to the visual transit because he regarded it as the backup, and that beforehand he would have said he would stop.

Conditions did not prevent a stop. There was no other traffic, the water was flat, the wind was offshore at 8 kt, and the vessel's trials figure for a crash stop from 8 kt was 92 m. The vessel crossed the channel's western edge at about 18:09, 114 seconds after the freeze. A stop at any point in that period would have left the vessel in the channel rather than on the bank. It is extremely likely that continuing the transit contributed to the vessel leaving the channel and to the late detection of shoaling.

The master was also without a second person on the bridge. The deckhand had been sent to collect fares at 18:04:14, which was the normal practice on the route, and OPS-3 did not address it either way.

This action did not occur in isolation. It followed from a passage plan that gave no hold point or abort criterion for loss of the primary means of fixing, which is discussed under Risk Controls. The visual backup he turned to depended on a beacon that could not be seen from that stretch of the channel, as discussed under Local Conditions.

Engines stopped and no attempt made to refloat on a falling tide

It is virtually certain that the master stopped the engines and did not attempt to refloat. The engines were stopped at 18:12:31, 27 seconds after contact, and were not restarted. Passengers were first addressed at 18:12:40, 36 seconds after the impact. At 18:13:04 the master's pan pan broadcast stated "I am not attempting to refloat" as a decision already made. He had the voids sounded, reported "approximately seventy five millimetres on the sounding rod" in the No. 2 starboard void at 18:17:05, and had it sounded every ten minutes after that. All 38 passengers were issued lifejackets and transferred ashore by 19:26. In interview he gave his reasons as the falling tide, the unknown bottom, the 38 people aboard, and that the vessel could not sink while it was on the bank.

The survey later confirmed a 180 mm split in the starboard bottom plating in way of the No. 2 void, with ingress reaching approximately 260 mm by the time the vessel refloated at 00:20. The breach was in the compartment the vessel was resting on. Backing off into 3.2 m of water at night, with the ingress not yet assessed, would have taken a flooding compartment into deep water with the passengers still aboard. It is very likely that the decision reduced the risk to those on board. The same analysis that identifies the decision to continue after the plotter froze also recognises that the master's actions after contact were sound and were taken promptly.

Occurrence Events

Vessel left the dredged channel and tracked onto Boolarong Bank

It is virtually certain that the vessel left the dredged channel and tracked onto Boolarong Bank. It is also virtually certain that this contributed to the grounding.

The track before the plotter froze. The AIS track placed the vessel 1 m west of the charted centreline abeam the No. 2 beacon at 18:05:30. It held the centreline to within 3 m until 18:07:06, the time at which the chart plotter event log recorded the display freeze.

The departure from the channel. From 18:07:06 the heading increased from 195 to 200 degrees over about a minute and was then held. The vessel moved west at an average of 12 m per minute. It reached the western edge of the 40 m wide channel at about 18:09:00 and was 39 m west of the centreline at 18:10:02. It was over Boolarong Bank from about 18:11:00.

The grounding position. The surveyed grounding position was 74 m west of the centreline. This is 54 m outside the channel and 34 m west of the No. 4 beacon. The furrows in the bed ran 11 m from the point of first contact on a heading of 197 degrees, which agreed with the recorded track.

How the vessel left the channel. The shape of the track showed this. The recorded rate of turn peaked at 2.1 degrees per minute at 18:07:40 and returned to zero by 18:08:50. That was a heading change applied and then held, not the gradual drift a current or wind would produce. The wind was offshore at 8 kt and there was no cross-set in the channel at that state of the tide.

Timing against the beacon search. The departure from the centreline began within a minute of the plotter freezing. It continued while the master was looking for the No. 4 beacon. When the

bridge audio recorded him saying "There's four. Keep her over" at 18:10:02, the vessel was already 39 m west of the centreline and outside the dredged channel.

Link to the grounding. The channel was dredged to 3.2 m. The grounding followed directly and only from the vessel being outside it.

Where the related matters are discussed. The conditions that brought the vessel there are discussed elsewhere in this analysis:

- The No. 4 beacon not being visible from the channel over the half mile before the vessel left it is discussed under Local Conditions.
- The continuation of the transit at eight knots after the plotter froze is discussed under Individual Actions.
- Whether new foreshore floodlighting behind the beacon reduced the master's ability to pick it up could not be determined. It is discussed as an unresolved matter under Local Conditions.

Shoaling not detected until sixteen seconds before contact

It is virtually certain that the shoaling was not detected until 16 seconds before contact. It is extremely likely that this contributed to the grounding.

The alarm and the response. The echo sounder alarm log recorded that the "Shallow alarm set to 0.9 m under the keel" was "the value written into the setup when the unit was commissioned in 2019 and it has not been changed." The alarm sounded once on the voyage, at 18:11:48, and the bridge audio recorded the tone beginning at that second. Both throttles were moved to astern together 4 seconds later, at 18:11:52. Contact followed at 18:12:04. The alarm log, the bridge audio and the recorded speed agreed to the second.

Warning time against stopping distance. At 7.1 kt the vessel covered 58 m in the 16 seconds between the alarm and contact. The vessel's own trials figure for a crash stop from 8 kt was 92 m. Astern thrust was developing at the moment of contact but had not reached full, and the vessel struck the bank at 6.8 kt. The surveyed gradient across that section of the bank was about 1 in 70. On that gradient, the alarm log recorded that the alarm "could not have sounded appreciably earlier at that setting."

What this shows. The echo sounder did exactly what it was set to do. The shortfall lay in the combination of the alarm setting and the transit speed, not in the equipment. On a gradient of about 1 in 70 at about 8 kt, a 0.9 m alarm gave less warning than the distance the vessel needed to stop. Either a setting that gave a substantially longer warning, or a transit speed at which 16 seconds was enough to stop, would have prevented the contact.

Steering or propulsion failure caused the vessel to sheer off track

A steering or propulsion failure was considered as an explanation for the vessel leaving the channel and was found very unlikely. The engine recorder showed both engines at 1,780 and 1,784 rpm with no alarm or throttle movement until 18:11:52. Steering hydraulic pressure held at 96 to 99 bar within its normal band, and rudder angle followed helm demand within 1.5 degrees. The rate of turn showed a heading applied and held rather than a sheer. The steering system was removed and function tested on 12/07/2026 and every component was serviceable, with no sign of a pre-existing defect.

Findings

From the evidence available, the following findings are made with respect to the grounding of MV Tarrakin Star on Boolarong Bank, Weerona Channel, Kanooka Heads, on 07/07/2026. These findings should not be read as apportioning blame or liability to any particular organisation or individual.

A finding stated without a likelihood term was established as fact.

Contributing factors

- The winter timetable moved the last service, TF18, into darkness under change record CHG-2026-014 without reopening route risk assessment RRA-07, which restricted the channel to daylight transits. It is likely that this contributed by leaving the channel without the night passage plan RRA-07 required, although a review might have produced a plan that said no more than the daylight one. (Safety Issue)
- It is virtually certain that there was no night passage plan for the Weerona Channel, and no hold point if the primary means of fixing was lost. It is very likely that this absence contributed to the master continuing the transit after the plotter froze rather than stopping in the channel. (Safety Issue)
- The change-of-master defect brief required by OPS-9 was verbal, with nothing checking that what was said got written down. This likely contributed by allowing three or four freezes of the vessel's primary means of position fixing over the preceding year to go unrecorded, so that firmware addressing the fault, issued in 11/2024, was never applied. A defect book entry made on the day would have been reviewed at the end of that week, however, and would not by itself have stopped TF18 sailing. (Safety Issue)
- The plotter fault was passed on verbally at the change of master at about 13:05 and was not entered in the defect book. It is likely that this contributed by keeping a fault the manufacturer had already addressed out of sight of anyone ashore who could act on it, although a single entry on a weekly review cycle would not by itself have stopped TF18 that evening.
- At 18:07:06, during the channel transit, the chart plotter display froze and did not recover. This very likely contributed by removing the primary means of position fixing named in the passage plan, and the vessel left the centreline within two minutes.
- After the plotter froze, the transit continued at eight knots rather than stopping in the channel. This very likely contributed to the grounding, as a stop at any point in the 114 seconds before the vessel crossed the channel's western edge would have left it in the channel rather than on the bank.
- Aids to navigation on the Weerona Channel carried no availability standard and no duty to tell users anything, because the duties at section 4.2 of WAN-2 apply only to declared shipping channels. On balance this likely contributed, since the notification duty was the one mechanism that would have put the No. 4 beacon's range change in front of the operator. The evidence for this was weaker than for the other contributing factors, however, because nothing in the code prevented the Authority from notifying operators, trimming the mangroves or measuring the light. (Safety Issue)
- In 11/2025 the No. 4 beacon was converted to a solar lantern, halving its nominal range from 6 to 3 nautical miles, and no operator was told. It is likely that this contributed by leaving the master working to a passage plan that still recorded the beacon at 6 M. Over the stretch where he searched for it, however, the light was screened entirely, and a brighter light would not have been visible there. (Safety Issue)

- Clearing of the beacon's sightline through the mangrove canopy on Weerona Islet was deferred as unfunded at three successive works rounds. This very likely contributed by leaving the beacon screened from the northern approach. (Safety Issue)
- It is very likely that the No. 4 beacon was not visible from the channel between 0.84 nm and 0.34 nm north of it, over the half mile before the vessel left the channel, based on a measurement made eleven days after the occurrence. It is very likely that this contributed to the vessel leaving the dredged channel, because with the plotter frozen the beacon was the only means of fixing available.
- From 18:07:06 the vessel's heading increased from 195 to 200 degrees and was held, and the vessel left the dredged channel and tracked onto Boolarong Bank. This contributed directly to the grounding.
- Shoaling was not detected until sixteen seconds before contact, because the echo sounder shallow alarm remained at its 2019 commissioning setting of 0.9 m under the keel. In that time the vessel covered 58 m against a crash stop distance from 8 kt of 92 m, and it is very likely that this contributed to the grounding.

Other factors that increased risk

- The forward saloon lifejacket locker on MV Tarrakin Star could not be reached from 22 of the 34 seats in the saloon it serves without passing behind a kiosk counter. The counter had been fitted in a 2019 refit under a minor modification notice that did not trigger re-survey of the safety equipment arrangement, and the same counter was fitted to the operator's two other 24 m vessels. It is very likely that this increased risk to passengers in any emergency requiring them to equip themselves without crew help. (Safety Issue)
- On day nine of twelve, the master was working a split duty across a spread of 13 hours 40 minutes that ended with the darkness transit of TF18. This was inside every applicable limit, and it is a pattern the winter roster places on every master on the route. It is likely that this rostering increased risk on the route's night channel transits.

Other findings

- The Port Authority gauge recorded 0.91 m at 18:12 against a predicted 0.94 m. It is very unlikely that a water level below prediction reduced the depth over Boolarong Bank.
- Recorded engine, steering and rudder data, and a full examination of the steering system after the occurrence, showed no defect. It is very unlikely that a steering or propulsion failure caused the vessel to sheer off track.
- After the grounding, the master stopped both engines and made no attempt to refloat on a falling tide with the vessel holed. This very likely reduced the risk to the 38 passengers on board.

Safety issues and actions

Aids to navigation on this waterway carry no availability standard and no duty to tell users anything

The Weerona Channel's beacons carried no availability standard and no duty to tell users anything. Under section 4.2 of the Waterways Aids to Navigation Code WAN-2, three duties apply only to aids on a declared shipping channel:

- achieving an availability of not less than 99.0 per cent
- recording availability
- notifying a change to an aid's character, position or range by notice to mariners

The Weerona Channel is not a declared shipping channel, so its beacons fell under section 4.3. That section requires only that the responsible authority can demonstrate an inspection interval it has set itself. The services agreement under which the beacons were maintained imposed no duty to tell any waterway user of a defect, a deferral or a change. The Kanooka Heads Port Authority was not in breach of either instrument.

This mattered because the notification duty was the one mechanism that would have told the operator about the No. 4 beacon's halved nominal range. The operator's passage plan still recorded the earlier figure on the night of the grounding. It is likely that the absence of these duties contributed to the occurrence. However, the evidence for this was weaker than for the other contributing factors. Nothing in the code prevented the Authority from notifying operators, trimming the mangroves or measuring the light. An availability target would also not have detected a lantern that was lit but could not be seen. More broadly, the code's threshold left a 40 m wide channel, used by a 96-passenger ferry, with beacons that nobody was required to measure, record or report on.

No safety action has been recorded in response to this safety issue.

Beacon converted to solar, nominal range halved, and no operator was told

In 11/2025, under the Council's waterways energy programme, the No. 4 beacon was converted from a mains supply to a solar lantern. The new lantern is rated at a nominal 3 nautical miles, against 6 for the lantern it replaced, and no operator using the channel was told of the change. The light's character, Fl G 2.5s, did not change. A notification was lodged with the hydrographic service on 04/02/2026, but the chart correction was still pending on the day of the grounding. No local notice to mariners was issued. The luminous range measured from the port workboat on 18/07/2026 was 2.6 nm.

This issue matters because operators were left navigating to written plans that described a light more than twice as strong as the one in place. Tarrakin Ferries' passage plan appendix, issue 2 and current on the night, recorded the No. 4 beacon at 6 M, eight months after the lantern had been replaced. It is likely that this contributed to the beacon not being visible from the channel. However, over the stretch where the master searched for it, the light was screened entirely, and a stronger light would not have been seen there.

On 22/07/2026, local notice to waterway users LNM-2026-14 was issued, giving the beacon's solar conversion and its 3 M nominal range. The Port Authority's aids to navigation register was made available to every commercial operator using the channel. Tarrakin Ferries has reissued the

Weerona Channel passage plan appendix as issue 3 with the corrected range. This action is held by the Aids to Navigation Officer, Kanooka Shire Council, and is in progress.

Clearing of the beacon's sightline deferred unfunded for three works rounds

Clearing of the No. 4 beacon's sightline through the mangrove canopy on Weerona Islet was deferred as unfunded at three successive works rounds. Work order WO-118, raised in 09/2023, recorded regrowth screening the beacon from the northern approach. It recommended trimming to 2 m on a three-yearly cycle at an estimated \$18,500. At the 2023, 2024 and 2025 rounds, each entry read "defer, unfunded, refer next round". By 03/2026, work order WO-176 recorded the sightline as "now fully closed on the northern approach". The islet is 240 m from the beacon and so fell outside the 50 m vegetation scope of schedule 2.4. As a result, the work competed for the general waterways vote rather than the aids to navigation vote. The canopy measured 2.1 m in the 2019 aerial survey and 5.8 m on 18/07/2026, against a lantern height of 4.2 m.

The issue mattered because the light was not dim or defective but absent from the water it served. The screened arc ran from 0.84 nm to 0.34 nm north of the beacon, and the vessel left the channel within it.

In response, the Harbour Master, Kanooka Heads, funded WO-176 from the 2026 waterways contingency. The canopy was trimmed to 2 m on 24/07/2026, which restored the sightline over the northern approach. A three-yearly trimming cycle has been programmed, and the sightline was re-measured from the inbound track on completion.

The investigation has also recommended that the aids to navigation services agreement replace the 50 m vegetation scope at schedule 2.4 with a sightline test. Under that test, work needed to keep an aid visible from the water it serves would be funded as an aid to navigation, whatever its distance from the structure.

Winter timetable moved the last service into darkness without reopening the route risk assessment

The operator's winter timetable took effect from 01/06/2026. Under change record CHG-2026-014, approved by the Operations Manager on 22/04/2026, it moved service TF18 from 16:15 to 17:35. This placed the last service of the day in darkness without reopening route risk assessment RRA-07.

RRA-07 issue 2 (03/2023), the current assessment for the Weerona Channel, carried the restriction "daylight transits only". It required the General Manager Operations' approval and a specific passage plan for any night transit. The change was assessed as "timetable change only", with "no new hazard identified". The question of whether a route risk assessment review was required was answered "No". It is virtually certain that RRA-07 was not reopened, that no night passage plan was written, and that the General Manager Operations' approval was neither sought nor given.

The issue was significant because the change put a 96-passenger vessel through a 40 m channel in darkness 31 times before the occurrence, contrary to a restriction that was still in force. The change process asked whether the change was new, not whether the existing assessment still held. A review against a night operation would have reached the unlit reach between the No. 2 and No. 4 beacons, which RRA-07 already named as a hazard. It is likely that the absence of that review contributed by leaving the channel without the night passage plan the assessment required.

No safety action has been recorded in response to this safety issue.

No night passage plan for the channel, and no hold point if the primary means of fixing is lost

There was no night passage plan for the Weerona Channel, and nothing in the operator's procedures told a master what to do if the primary means of position fixing was lost during the transit. Appendix A of OPS-3 named the chart plotter as the primary means of fixing. It named the visual transit of the No. 4 beacon with the town wharf leading lights as the secondary means. The plan was written for the daylight transit and named no hold point or abort criterion. OPS-3 3.6 required a reduction of speed and consideration of anchoring only in reduced visibility, which did not apply on the night. It is virtually certain that this gap existed. It is very likely that it contributed to the master continuing the transit at eight knots after the plotter froze.

The issue matters because the passage plan was the operator's own defence against the sequence that occurred, and no version of it had been written for darkness. The master had 114 seconds between the freeze and the vessel crossing the channel's western edge, on flat water with no other traffic. A named hold point would have been available throughout that time. Both masters interviewed said independently that stopping was the right response and that nothing in the procedures asked for it.

Proposed safety action

The investigation recommended that Tarrakin Ferries write a night passage plan for the Weerona Channel before any further darkness transit. It also recommended that the plan name a hold point, so that loss of the primary means of position fixing inside the channel requires the way to be taken off the vessel until its position is re-established by another means. A response from Tarrakin Ferries was due by 30/09/2026.

[NOT ESTABLISHED: Tarrakin Ferries' response to the recommendation, if one has been received]

Change-of-master defect brief is verbal, with nothing checking that what is said gets written down

Under Operations Manual OPS-9, the change-of-master defect brief was verbal, and nothing checked that a defect mentioned in it was entered in the defect book. Section 9.1 required immediate reporting only of a defect rendering the vessel unseaworthy or affecting a statutory item. The chart plotter was not a statutory item on a class 1D vessel on this route, although it was the primary means of position fixing named in the passage plan. Section 9.2 placed all other defects in the defect book at the end of the shift in which they were found, which gave no trigger for a fault that came and went. Section 9.3 required the outgoing master to brief the relieving master but required nothing to be written down, and nothing of the relieving master afterwards.

Both masters independently described the same understanding. The book was for what was broken at handover, and what had come good was spoken about. This practice explains how the plotter fault went unrecorded. It does not excuse the omission.

The outgoing master believed the plotter had frozen three or four times on the vessel in the preceding year, and none of these freezes had been written up. The maintenance system held no plotter entry for 2026, and firmware addressing the fault, issued in 11/2024, was never applied. It is likely that this contributed to the occurrence. However, an entry made at 13:05 on the day would have been reviewed at the end of that week, and TF18 would still have sailed. On that reading, the rule is a finding about the fleet rather than a cause of this grounding.

No safety action has been recorded in response to this safety issue.

Forward saloon lifejacket locker not reachable from most of the saloon it serves

On MV Tarrakin Star, passengers could not reach the forward saloon lifejacket locker from most of the saloon it serves. A survey of the vessel on 16/07/2026 found two problems:

- From 22 of the 34 forward saloon seats, the locker face could not be reached without passing behind the kiosk counter.
- With people standing at the counter, the only route was an inboard gap of 0.6 m.

The muster placard directed passengers to a locker they could not see from their seats. While issuing lifejackets after the grounding, the deckhand had to ask three passengers to move away from the counter and pass behind it to reach the locker.

The arrangement approved at build in 2011 had an open bench in that position, with the locker face clear. The counter was fitted in a 2019 interior refit under a minor modification notice, which did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels.

The arrangement had no effect on the grounding or its outcome. The vessel was aground, upright, stable and not sinking, and all 38 passengers were in lifejackets within about eight minutes. The circumstances could have been different, however. In a fire, a capsize risk or a night evacuation into the water, or with a deckhand injured in the impact, passengers who could not reach the lifejackets without crew help would have had to wait for them. It is very likely that the arrangement increased risk to passengers. The underlying gap was in the modification route, which allowed an interior refit to change a safety equipment arrangement without asking whether it had done so.

No safety action has been recorded in respect of this issue.

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